WASP-10b

R-0841

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_180926` · created 2026-07-28T19:11:11+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458387.676 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.016
Transit depth (Rp/Rs)^2	2.94 +/- 0.55 [%]
Orbital Inclination (inc)	87.02 +/- 0.98
Airmass coefficient 1 (a1)	0.911 +/- 0.017
Airmass coefficient 2 (a2)	0.074 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#2 - [315, 248]
Optimal Aperture	2.31
Optimal Annulus	11.47
Transit Duration (day)	0.081 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.016 vs 0.1592 (deviation 0.65σ)
Duration fit vs published	0.081 d vs 0.0946 d (deviation 1.09σ)
Mid-time O–C	-9.5 min
Depth SNR	9.4
Residual scatter	1.89 %

Light curve

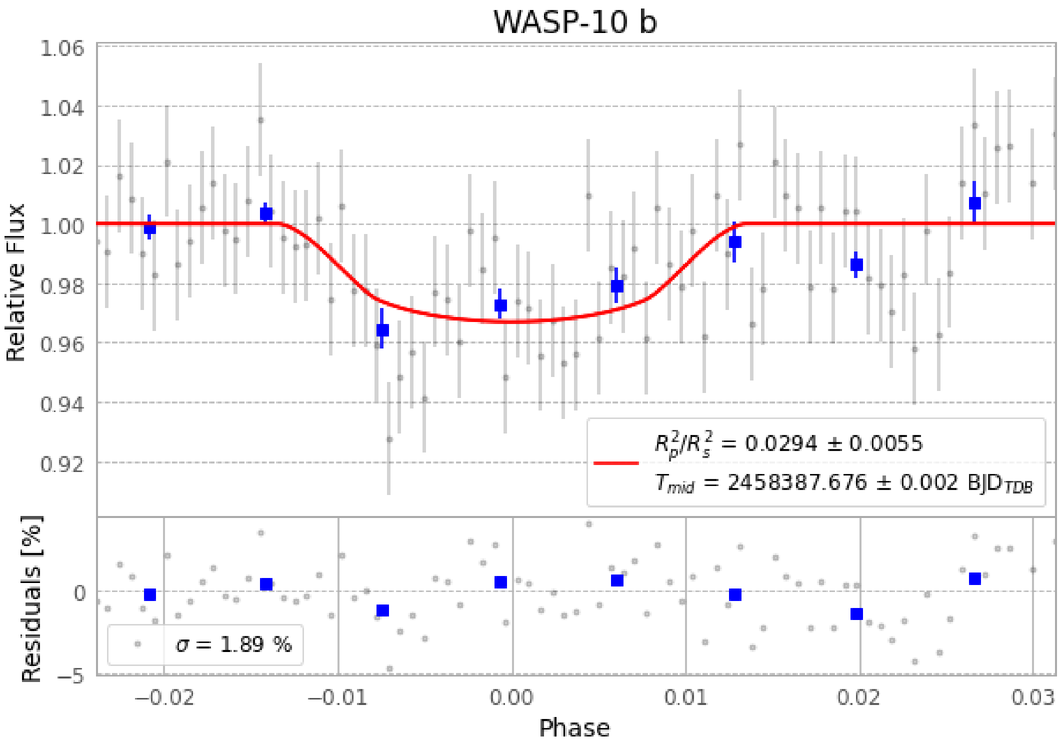

Figure 1 — FinalLightCurve_WASP-10 b_2018-09-25.png (SHA-256 2964ce1df0aa33ad…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 175], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 cf456b21c71bcdbf…).
`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

83 FITS frames (55 MB), `WASP-10180926021812.FITS` → `WASP-10180926062412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-180926.log (9d21d7992a8b…), FinalLightCurve_WASP-10 b_2018-09-25.png (2964ce1df0aa…), FinalLightCurve_WASP-10 b_2018-09-25.csv (7e56aa7bdf24…)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)